

Student Name: Anjali Sharma

Roll No: 12CSA001

Subject: Mathematics

School / Institution:	Delhi Public School, R.K. Puram
Student Name:	Anjali Sharma
Roll Number:	12CSA001
Class:	XII-A
Subject:	Mathematics
Barcode:	BAR101
Date of Examination:	15 March 2025
Session:	2024-25

I hereby declare that the answers written in this booklet are my own. I have not used any unfair means during the examination.

Candidate's Signature: _____

Invigilator's Signature: _____

Room No: _____

SECTION A — Very Short Answer (1 mark each)

Q1. Find the principal value of $\sin^{-1}(1/2)$.

[1 mark]

$\sin^{-1}(1/2) = \pi/6$

Because $\sin(\pi/6) = 1/2$ and $\pi/6$ lies in the principal value branch $[-\pi/2, \pi/2]$.

Therefore, the answer is $\pi/6$.

Marks	
Awarded	
/1	
Initials	

Q2. Write the order and degree of: $(d^2y/dx^2) + 3(dy/dx) + 2y = 0$

[1 mark]

Order = 2 (highest derivative is d^2y/dx^2)

Degree = 1 (power of highest derivative is 1)

The differential equation is linear with constant coefficients.

Student Name:Anjali Sharma

Roll No:12CSA001

Subject:Mathematics

Marks	
Awarded	
/1	
Initials	

Q3. If A is a square matrix of order 3 and $|A| = 5$, find $|2A|$. [1 mark]

$|2A| = 2^3 |A| = 8 \times 5 = 40$

This is because for an $n \times n$ matrix, $|kA| = k^n |A|$.

Here $n=3$, $k=2$, so $|2A| = 2^3 \times 5 = 40$.

—

—

—

—

—

Marks	
Awarded	
/1	
Initials	

Q4. Evaluate: $\int e^x (\sin x + \cos x) \, dx$ [1 mark]

$\int e^x (\sin x + \cos x) \, dx = e^x \sin x + C$

Using the formula $\int e^x [f(x) + f'(x)] \, dx = e^x f(x) + C$

Here $f(x) = \sin x$, $f'(x) = \cos x$. So the answer is $e^x \sin x + C$.

—

—

—

—

—

Marks	
Awarded	
/1	
Initials	

SECTION B — Short Answer (2 marks each)

Q5. Prove that the function $f(x) = x^2$ is strictly increasing on $(0, \infty)$. [2 marks]

Let $x_1, x_2 \in (0, \infty)$ such that $x_1 < x_2$

Then $x_1^2 < x_2^2$ (since both are positive)

$\Rightarrow f(x_1) < f(x_2)$

Therefore $f(x) = x^2$ is strictly increasing on $(0, \infty)$. Hence proved.

—

Student Name: Anjali Sharma

Roll No: 12CSA001

Subject: Mathematics

Marks	
Awarded	
/2	
Initials	

Q6. Find the area bounded by the curve $y = x^2$ and the lines $x = 0$ and $x = 2$.

[2 marks]

$$\text{Area} = \int_{-2}^2 x^2 dx = [x^3/3]_{-2}^2$$

$$= (2^3/3) - (0^3/3)$$

$$= 8/3 - 0$$

$$= \frac{8}{3} \text{ square units}$$

The area enclosed between the parabola and the x-axis from $x=0$ to $x=2$ is $\frac{8}{3}$ sq. units.

Marks	
Awarded	
/2	
Initials	

Q7. If $y = (\sin x + \cos x)^2$, find dy/dx .

[2 marks]

$$y = (\sin x + \cos x)^2$$

Using chain rule: $\frac{dy}{dx} = 2(\sin x + \cos x) \times \frac{d}{dx}(\sin x + \cos x)$

$$= 2(\sin x + \cos x)(\cos x - \sin x)$$

$$= 2(\cos^2 x - \sin^2 x)$$

$$= 2\cos 2x$$

Marks	
Awarded	
/2	
Initials	

Student Name: Anjali Sharma

Roll No: 12CSA001

Subject: Mathematics

SECTION C — Long Answer I (4 marks each)

Q8. Using matrices, solve the system: $2x + y = 5$, $x + 3y = 7$

[4 marks]

Matrix form: $AX = B$ where $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$, $B = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$ Finding A^{-1} :

$$|A| = (2 \times 3) - (1 \times 1) = 6 - 1 = 5$$

$$\text{Adjoint of } A = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$$

$$A^{-1} = (1/5) \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$$

$$X = A^{-1} B = (1/5) \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$$= (1/5) [(15-7), (-5+14)]$$

$$= (1/5) [8, 9]$$

$$x = 8/5, y = 9/5$$

$$\text{Verification: } 2(8/5) + 9/5 = 16/5 + 9/5 = 25/5 = 5 \checkmark$$

Marks	
Awarded	
/4	
Initials	

SECTION D — Long Answer II (6 marks each)

Q9. Find the area of the region bounded by the ellipse $x^2/4 + y^2/9 = 1$

[6 marks]

The given ellipse has $a = 2$, $b = 3$ By symmetry, Area = $4 \times \int_{-2}^2 y \, dx$ where $y = (3/2)\sqrt{4-x^2}$

$$\text{Area} = 4 \times (3/2) \int_{-2}^2 \sqrt{4-x^2} \, dx$$

$$\text{Let } x = 2\sin\theta, \, dx = 2\cos\theta \, d\theta$$

$$\text{When } x=0, \theta=0; \text{ when } x=2, \theta=\pi/2$$

$$\sqrt{4-x^2} = \sqrt{4-4\sin^2\theta} = 2\cos\theta$$

$$= 4 \times (3/2) \int_0^{\pi/2} 2\cos\theta \times 2\cos\theta \, d\theta$$

$$= 12 \int_0^{\pi/2} 4\cos^2\theta \, d\theta \leftarrow (\text{error here, correcting})$$

$$= 6 \int_0^{\pi/2} 4\cos^2\theta \, d\theta$$

Student Name: Anjali Sharma

Roll No: 12CSA001

Subject: Mathematics

$$= 6 \int_0^{\pi/2} 2(1+\cos 2\theta) \, d\theta$$

$$= 12 [\theta + \sin 2\theta/2] \Big|_0^{\pi/2}$$

$$= 12 [\pi/2 + 0 - 0]$$

$$= 6\pi \text{ square units}$$

The area of the ellipse = 6π sq. units

(Formula check: $\text{Area} = \pi ab = \pi \times 2 \times 3 = 6\pi$ ✓)

Marks	
Awarded	
/6	
Initials	

Student Name:Anjali Sharma

Roll No:12CSA001

Subject:Mathematics

*** END OF ANSWER BOOKLET ***

MARKS SUMMARY (For Examiner Use Only)

Section	Max Marks	Marks Obtained	Examiner Initials
Section A (1 mark × 4)	4		
Section B (2 marks × 3)	6		
Section C (4-5 marks)	varies		
Section D (5-6 marks)	varies		
TOTAL	100		